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PAPER_1103: LQG Buoyancy Sector Lagrangian Variation with Spin-Foam Vertex Coupling

Abstract

We construct the Lagrangian for the LQG buoyancy sector, coupling the spin-foam vertex amplitude $A_v(j, i_e)$ to the UQFF buoyancy scalar field ϕ_{LQG} and the 1.25 THz phonon mode. The Euler-Lagrange equation $\delta S / \delta \phi_{\text{LQG}} = 0$ yields an equation of motion with spin-foam source terms and phonon-induced effective mass corrections.

1. Lagrangian Density

$$\mathcal{L}_{\text{LQG}} = \frac{1}{2}(\partial_\mu \phi_{\text{LQG}})^2 - V(\phi_{\text{LQG}}) + \lambda_{\text{spin}} \sum_v A_v(j, i_e) \cdot \phi_{\text{LQG}} + g_{\text{phonon}} \Phi_{1.25 \text{ THz}} \phi_{\text{LQG}}^2$$

Terms

Term	Physical origin
$\frac{1}{2}(\partial_\mu \phi)^2$	Kinetic energy of the buoyancy field
$V(\phi) = \frac{1}{2}\omega_\phi^2 \phi^2$	Harmonic potential, $\omega_\phi = m_\phi c^2 / \hbar$
$\lambda_{\text{spin}} \sum_v A_v \cdot \phi$	Spin-foam vertex coupling (EPRL/FK)
$g_{\text{phonon}} \Phi \phi^2$	Phonon-buoyancy self-interaction

2. Spin-Foam Vertex Amplitude

Using the Ponzano-Regge/EPRL approximation:

$$A_v(j) = (2j + 1) \cdot \exp(-\gamma j)$$

where $\gamma = 0.2375$ is the Barbero-Immirzi parameter. The sum over N_v vertices:

$$\sum_v A_v = N_v \cdot (2j_{\text{avg}} + 1) \cdot \exp(-\gamma j_{\text{avg}})$$

3. Equation of Motion

Varying $\delta S / \delta \phi_{\text{LQG}} = 0$:

$$\square \phi_{\text{LQG}} + \omega_\phi^2 \phi_{\text{LQG}} = \lambda_{\text{spin}} \sum_v A_v + 2g_{\text{phonon}} \Phi \phi_{\text{LQG}}$$

This is a **Klein-Gordon equation with spin-foam source and phonon-modified mass**:

$$\square\phi + m_{\text{eff}}^2\phi = J_{\text{spin-foam}}$$

where:

$$m_{\text{eff}}^2 = \omega_\phi^2 - 2g_{\text{phonon}}\Phi$$

$$J_{\text{spin-foam}} = \lambda_{\text{spin}} \sum_v A_v(j, i_e)$$

4. Tachyonic Instability Condition

If $2g_{\text{phonon}}\Phi > \omega_\phi^2$, the effective mass squared becomes negative, signaling a tachyonic instability. This triggers spontaneous symmetry breaking in the buoyancy sector, with the new vacuum:

$$\langle\phi\rangle = \pm\sqrt{\frac{J_{\text{spin-foam}}}{m_{\text{eff}}^2}}$$

5. Static Equilibrium

At static equilibrium ($\square\phi = 0$), the balance condition:

$$\omega_\phi^2\phi_0 = \lambda_{\text{spin}} \sum_v A_v + 2g_{\text{phonon}}\Phi\phi_0$$

The equation of motion residual R :

$$R = V'(\phi_0) - J_{\text{spin-foam}} - 2g_{\text{phonon}}\Phi\phi_0$$

with balance ratio $|R|/|V'(\phi_0)|$ indicating proximity to equilibrium.

6. Implementation

Calculator: `LQGBuoyancySectorLagrangianVariationCalculator` in `CondensedPhysics.py`

- Inputs: field amplitude ϕ_0 , mass m_ϕ , couplings λ_{spin} and g_{phonon} , vertex count N_v , average spin j_{avg}
- Outputs: Lagrangian density, spin-foam and phonon source terms, equation of motion residual, effective mass, tachyonic flag

7. Conclusion

The LQG buoyancy sector Lagrangian provides the variational framework for deriving the dynamics of the UQFF buoyancy field in the spin-foam formalism. The phonon-induced effective mass correction and the spin-foam source term together determine the equilibrium configuration and stability of the buoyancy sector.

References

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